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Part 1: Linear Regression

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Regression Linear

Introduction

Healthcare and insurance industries rely on data for decision-making through predictive analytics. They use data-driven models to estimate medical costs and set premiums for individuals depending on different characteristics. This research focuses on predicting medical aid costs using a dataset that includes lifestyle and demographic variables such as age, sex, BMI, number of children, smoking status, and region. The used in this research was sourced from Kaggle (Mirichoi, 2018)

It is important to understand factors that influence medical aid costs. Medical aid costs may vary from one individual to the other based on different things. People who live an unhealthy lifestyle usually pay more price because they can easily get sick. In the exploratory data analysis (EDA), we will see the relationship between the independent variables and the dependent variable (charges). Recent research shows that data analytics plays a huge role in improving healthcare decision-making and cost prediction (Wang et al., 2019)

To achieve the research objective, we developed a linear regression model to predict insurance charges based on selected features. Linear regression is used for predicting continuous outcomes due to its simplicity and interpretability (James et al., 2021). Before the model training, we applied the data preprocessing techniques, converted categorical variables into numerical format using one-hot encoding. The dataset was then split into training and testing sets to evaluate the model performance on unseen data.

We used metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE),Root Mean Squared Error (RMSE), and the coefficient of determination (R^2) to evaluate the model. This research shows that machine learning techniques can be applied in real-world insurance data to make predictions and draw insights. The findings help us understand factors influencing insurance charges and also help us set premium costs within the insurance industry.

Evaluate dataset suitability for linear regression

Continuous dependent variable

Linear regression is specifically designed for modelling continuous response variables (James et al., 2021). For the dataset to be suitable for linear regression, the dependent

variable (charges) should be continuous (numeric). In our case the dependent variable is indeed continuous making our dataset suitable for linear regression. Figure 1 shows that charges are continuous and right skewed.

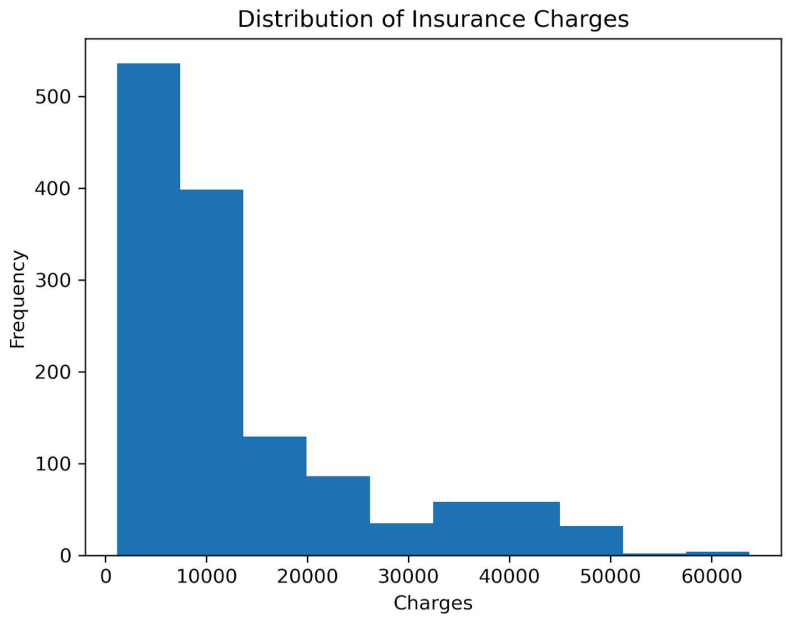
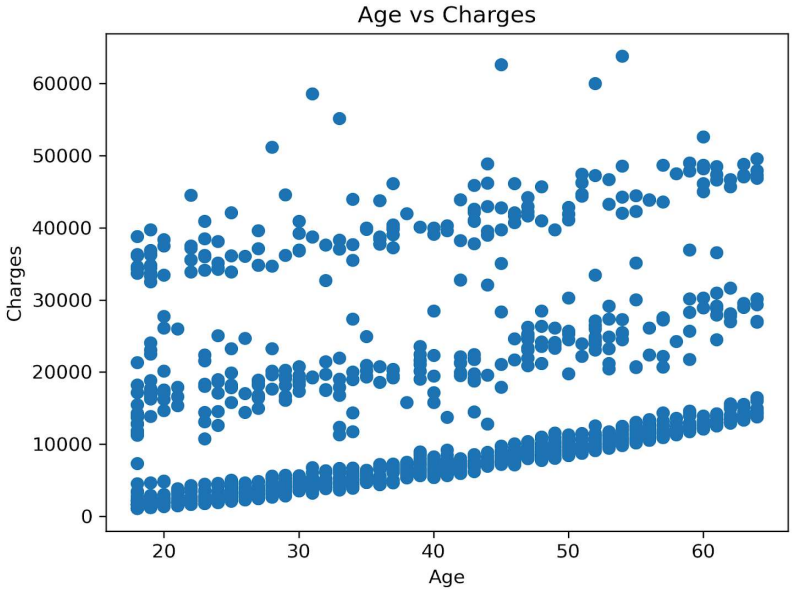


Figure 1: Histogram showing the distribution of insurance charges

Linearity between variables

There should be a linear relationship between independent variables (age, BMI, gender of the client, smoking status, number of children, and geographic region) and the dependent variable (charge). This can be shown using scatter plots. These plots help us determine

whether the relationship follows a straight line.



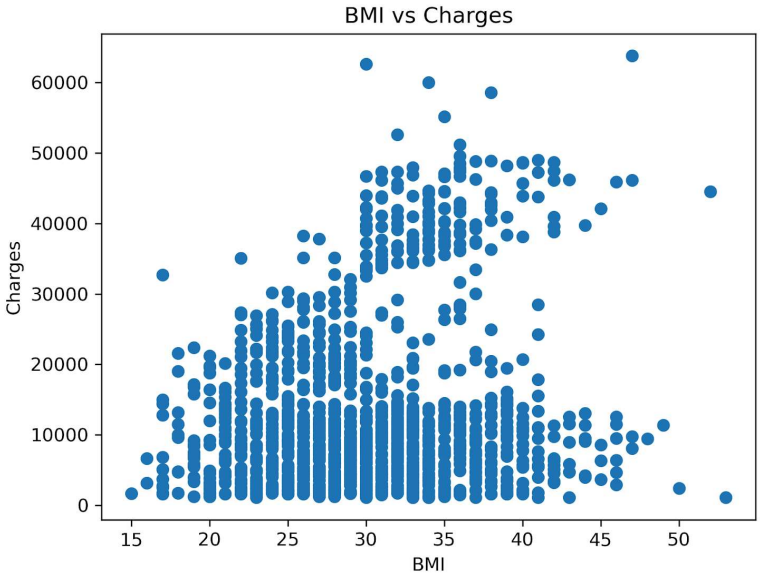


Figure 2: Scatter plot to determine linearity between variables.

Exploratory Data Analysis (EDA)

Exploratory data analysis (EDA) is an important step in data analysis that helps in understanding the structure of the dataset, identifying patterns, and detecting anomalies (GeeksforGeeks, 2025). Visualisation and summary statistics are part of EDA that draws insights that help us with feature selection and model development. According to Jame et al. (2021), EDA is essential for uncovering underlying trends and ensuring that the data is suitable for modeling. In this dataset, we checked for missing values, reviewed data types,

and generated summary statistics. We also visualised data using histogram, scatterplots, and box plots to reveal patterns and relationships between variables

1. Dataset structure

We checked the structure of the dataset to understand the types of variables and the composition of the data. There are both categorical variables (sex, smoker status, and region) and numerical variables (age, BMI, number of children, and charges). This tells us that we will need to perform preprocessing such as converting our categorical variable into numerical format before model training. Figure 3 shows our data structure.

```
df.describe()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

Figure 3: Dataset structure

2. Checking missing values

The dataset does not have any missing values. This means that the dataset is clean and ready for analysis or modeling.

3. Distribution of Insurance charges

We used a histogram to check the distribution of the target variable (insurance charges)

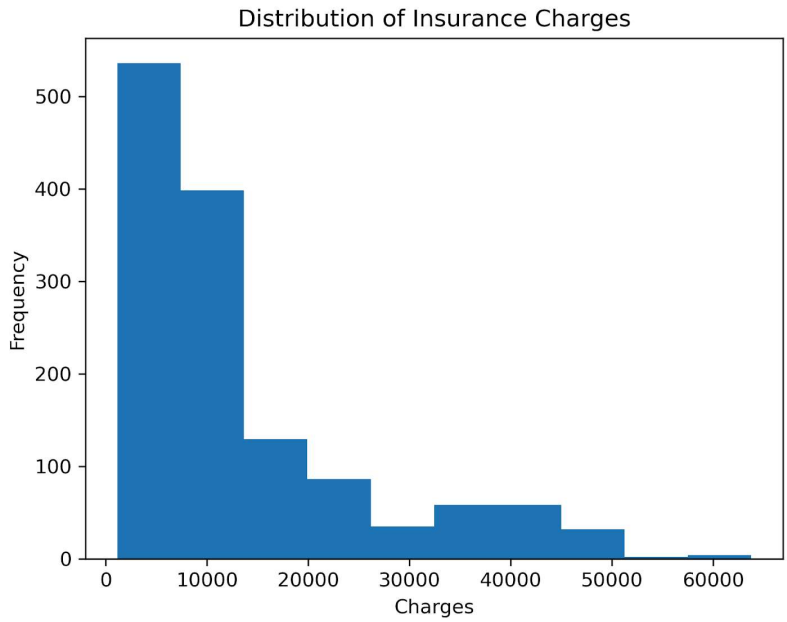


Figure 4: Distribution of Insurance Charges

The distribution of insurance charges is right skewed. This shows that a lot of individuals have lower charges while fewer individuals have higher charges.

4. Age vs Insurance Charges

We used a scatter plot to show the relationship between age and insurance charges.

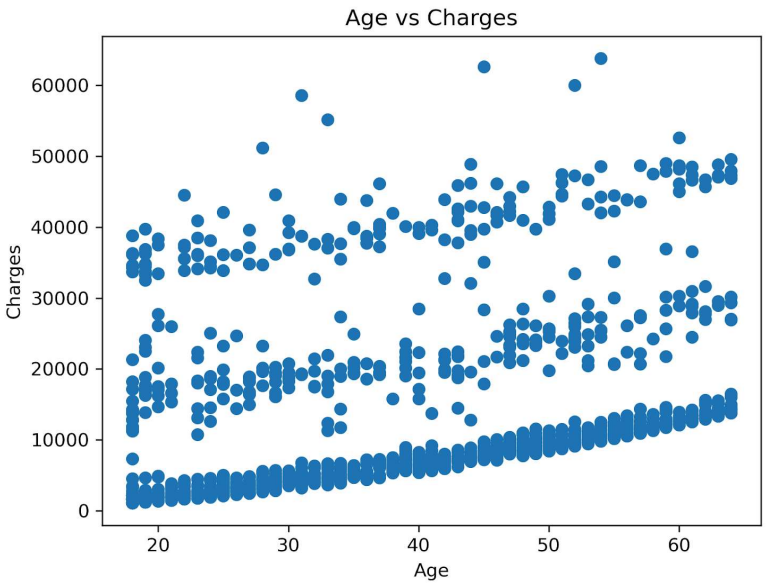


Figure 5: Scatter plot of age vs insurance charges

Figure 5 shows that younger individuals have less charges while older individuals have higher charges. This shows a positive relationship between age and insurance charges.

5. BMI vs Insurance Charges

We used a scatter plot to show the relationship between BMI and insurance charges.

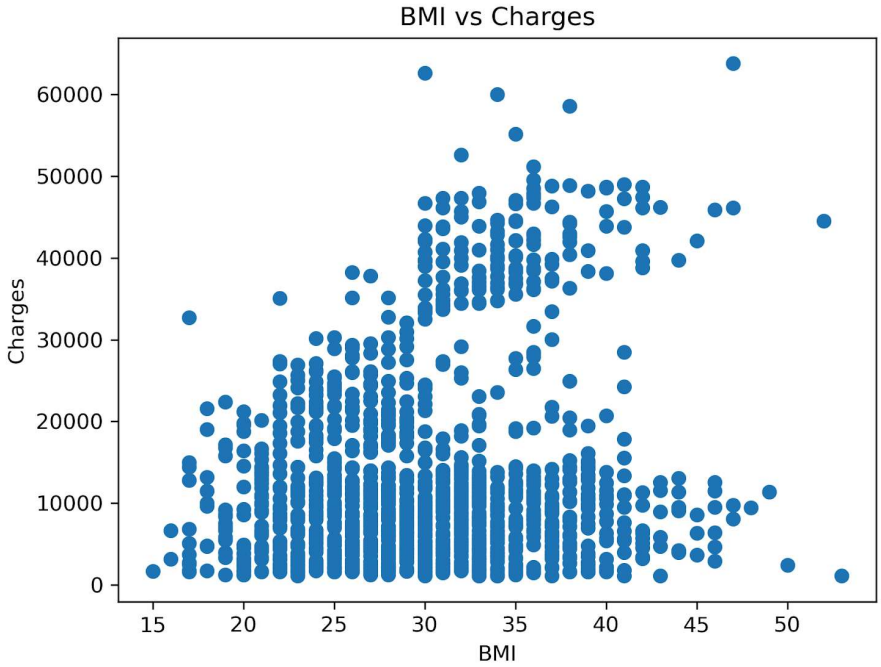


Figure 6: Scatter plot of BMI vs insurance charges

Individuals with higher BMI pay more insurance charges than individuals with lower BMI. BMI alone does not have much impact on charges, it needs other strong features to make a difference.

6. Impact of smoking status

We used a box plot to compare insurance charges between individuals who smoke and those who don't.

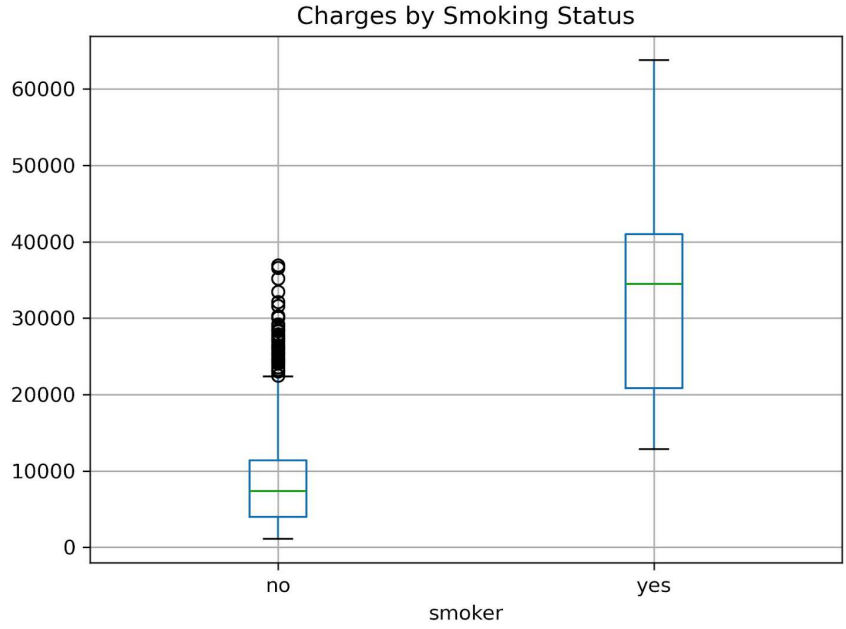


Figure 7: Boxplot of charges by smoking status

There is a huge difference in insurance charges between individuals who smoke and those who don't. Smokers have higher insurance charges while non-smokers have less insurance charges.

Summary of key findings

The EDA showed us these important insights:

- The dataset has both numerical and categorical variables.
- There are no missing values, meaning the data is clean and ready for analysis or modeling
- Insurance charges are right-skewed.
- Age has a positive relationship with charges.
- Smoking status has the most impact on charges.
- BMI has a moderate influence.

Feature Selection

Feature selection is the process of choosing only the most useful input features for a machine learning model. It helps improve model performance, reduces noise and makes results easier to understand (GeeksforGeeks, 2025). In this step, we converted our categorical variables into numeric format using encoding techniques because machine learning algorithms require numeric inputs (Snapwise, 2023). We also conducted a correlation analysis to determine the most relevant features for predicting medical charges.

1. Correlation analysis

Correlation analysis is widely used to identify important predictors in a dataset (Wang et al., 2019). We used a correlation matrix to evaluate the strength of the relationships between features and target variable.

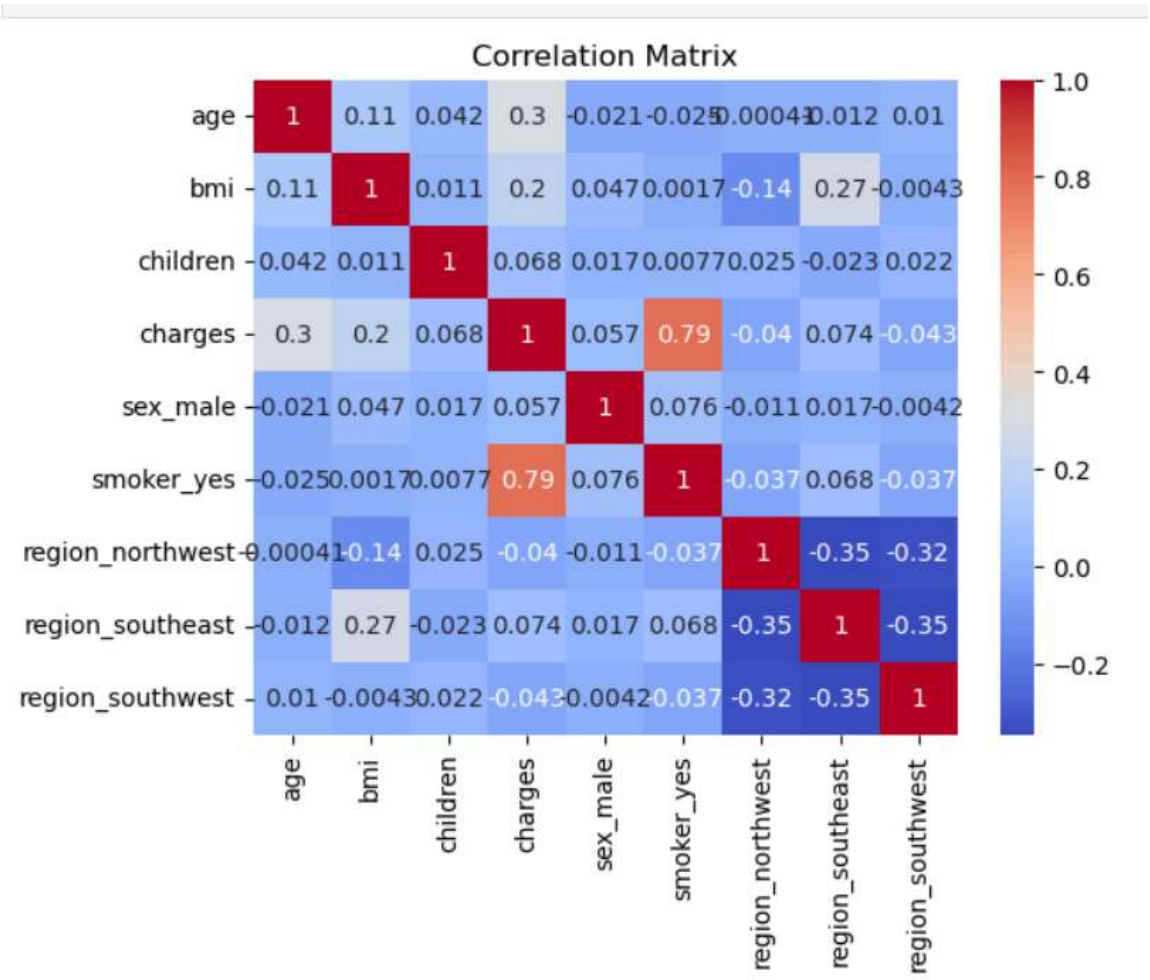


Figure 8: Correlation heatmap showing relationships between variables.

Figure 8 shows that smoking status has the strongest positive relationship with insurance charges. This means that people who smoke have higher medical costs. Age also shows a moderate positive correlation with charges, while BMI shows a weaker relationship. Other features such as region and children show low correlation with charges.

2. Conversion of categorical variables

The dataset contains categorical variables such as sex, smoker, and region. We converted these variables into numeric format using dummy variable encoding. This step is important because machine learning models require numeric inputs.

Model training

1. Splitting the dataset

In this step we split the dataset into a training and testing set. The training set is used to train the data while the testing set is used to assess the model performance on unseen data. We used 80/20 split where 80% of the data was used for training and 20% of the data was used for testing.

```
# Splitting the dataset
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

- X contains features
- Y contains the target variable

2. Training the model

We trained the model using the training dataset. In this step, the model learns the relationship between input features and medical charges. The model coefficient indicates the impact of each feature on this target variable. A positive coefficient shows that the variable increases medical charges while a negative coefficient shows a decrease.

3. Making predictions

After training the model, we used the test data to make predictions.

Interpret and evaluate model

In this step, we evaluated our model using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). These metrics provide a comprehensive understanding of the model's predictive capability (James et al., 2021). The model produced the following results:

- Mean Absolute Error (MAE): 4176.27
- Mean Squared Error (MSE): 33,566,439.74
- Root Mean Squared Error (RMSE): 5793.66
- R^2 Score: 0.784

The model achieved MAE of 4176.27, this indicates that the predicted values differ from the actual values by approximately 4176.27. RMSE of 5793, this is larger than MAE so it tells us larger error exists. The R^2 score of 0.784 which explains about 78.4% of the variation in medical charges. Overall, the model performs well but the difference between MAE and RMSE tells us that the model could be improved.

Conclusion

This research built a linear regression model to predict medical aid costs using demographic and lifestyle factors. Exploratory data analysis (EDA) showed us key patterns and the relationships between independent variables and dependent variable. Feature selection and data preprocessing ensured that the dataset was ready for modelling.

The model achieved an R^2 score of approximately 0.784, meaning a good predictive performance. Evaluation metrics such as MAE and RMSE show that there exist some prediction errors.

Overall, the results showed that linear regression is a productive method for predicting insurance charges, although the model could still be improved.

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